

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Scala For Data Science

Practical V

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Class: SYBsc IT	Subject: Scala For Data Science
Date of Assignment: 14/07/2026	Date/Time of Submission: 14/07/2026

AIM: Create a dense vector using Breeze and calculate its sum, mean, and dot product with another vector.

CODE:

```
1 ThisBuild / version := "0.1.0-SNAPSHOT"
2
3 ThisBuild / scalaVersion := "2.13.12"
4
5 libraryDependencies += "org.scalanlp" %% "breeze" % "2.1.0"
6 lazy val root = (project in file("."))
7   .settings(
8     ⚡ name := "PRACTICAL4"
9   )
10
```

```

1  import breeze.linalg._
2
3  ▶ object Demo extends App{
4      val A = DenseVector(1.0,2.0,3.0,4.0,5.0)
5      val t = sum(A)
6      val m = breeze.stats.mean(A)
7      val B = DenseVector(5.0,4.0,3.0,2.0,1.0)
8      val p = A dot B
9
10     println(s"1st Vector : $A")
11     print(s"Sum: $t ")
12     println(s"Mean: $m")
13     println(s"2nd Vector : $B")
14     println(s"Dot Product: $p")
15     💡 println("IQRA S021")
16 }
17

```

OUTPUT:

```

C:\Program Files\Java\jdk1.8.0_181\bin\java.exe" ...
1st Vector : DenseVector(1.0, 2.0, 3.0, 4.0, 5.0)
Sum: 15.0 Mean: 3.0
2nd Vector : DenseVector(5.0, 4.0, 3.0, 2.0, 1.0)
Dot Product: 35.0
IQRA S021

Process finished with exit code 0

```